

Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____

Maximums, Minimums, Increasing and Decreasing

Example (Read max/min and intervals):

$y = x^2 - 4$ has a **minimum** at $(0, -4)$. It **decreases** on $x < 0$ and **increases** on $x > 0$.

Directions: State whether there is a max or min, give the point, and the increasing/decreasing intervals.

1. $y = x^2 + 2$

3. $y = (x - 3)^2$

2. $y = -x^2 + 5$

4. $y = -(x + 1)^2 - 2$

Challenge Problems

5. $y = (x - 2)^2 - 9$ — give the minimum point and both intervals.

Answer Key and Procedure

Procedure

1. Find the vertex (turning point).
2. Decide max (opens down) or min (opens up).
3. Read increasing/decreasing on each side of the vertex.

Answers

1. Min $(0, 2)$; decreasing $x < 0$, increasing $x > 0$
2. Max $(0, 5)$; increasing $x < 0$, decreasing $x > 0$
3. Min $(3, 0)$; decreasing $x < 3$, increasing $x > 3$
4. Max $(-1, -2)$; increasing $x < -1$, decreasing $x > -1$
5. Min $(2, -9)$; decreasing $x < 2$, increasing $x > 2$